exp341 Track A: memorization vs coevolution in contacts-v1

Track A of #341 asks whether the next contacts-v1 model should cap how often each protein family is sampled (decision A in `research_plan.md`, section 5). A cap helps if the model recalls families and individual structures; it costs accuracy if the model learns couplings from family statistics.

Recommendation: no change to family weighting for the next model. Keep the corpus's current per-family sampling, and record decision A as inconclusive. The plan fixed a rule for each outcome before any result existed (section 1.2), and the results match neither row:

Plan's condition	"Cap per cluster" row	"Keep redundancy" row	Observed
A4: exposure advantage with many relatives left	persists		Persists: +0.040 R-precision at 64+ relatives
A3: accuracy vs surviving relatives	saturates at a small count	keeps rising	Flat from 0 relatives; control failed; breakpoint CI 2 to 435
A2: contact dependence	non-local	local	Local, and exposure makes it more local
A1: coupling map tracks MSA depth		yes	Not run

The plan reports a result between the rows as inconclusive, and a cap needs a value from A3 that the data do not give. The evidence behind this:

- **A4.** Training on a structure (together with its Foldseek cluster) adds 0.04 to 0.085 R-precision in every relative-count bin, including structures that keep 64 or more relatives in training. That is a memorized component, small next to the 0.56 R-precision exp232 reaches on the same structures without seeing them.
- **A3.** Its accuracy curve is flat from zero relatives, and its control set scores below the structures it was meant to bound, so it supplies no cap value.
- **A2.** Locality cannot separate recall from coupling in this model: exposure made each contact depend more on the residues at the contact, not less, which is the direction the plan assigned to coupling.

Two things would reopen the decision: an A3 control matched on whatever makes removed structures easier (no new GPU training needed), or a per-cluster cap ablation in training. Further A2 work (the fp32 rerun of the eval set) would not, since locality cannot decide A.

What ran

Part	Model	Proteins	Readout	State
A4	exp199 vs exp232	1,000 removed + 1,000 matched kept	R-precision, 4 seeds	Done

Part	Model	Proteins	Readout	State
A3	exp232	same panel	R-precision, 4 seeds	Done; control failed
A2, eval set	exp277	316 (eval-val, eval-denovo, 200 legacy-554)	contact Jacobian, bf16, 1 seed	Done; bf16-limited
A2, A4 panel	exp199 and exp232	100 matched pairs, both members <= 250 residues	contact Jacobian, fp32, 1 seed	Done
A1			sequence-section Jacobian	Not run

Models: `contacts-v1-exp199-cooldown-1.5B` (step 290400) trained on the structures #225 later removed; `contacts-v1-exp232-m2-p06-train-1.5B` (step 363000) never saw them; `contacts-v1-exp277-m2-p06-full-epoch-1.5B` is the registry default. All runs used one L40S (g6e.xlarge, us-east-2). The panel readouts took 1.39 h per model, A2 on the eval set 18.2 GPU-h, and A2 on the panel 20.9 GPU-h.

The held-out panel

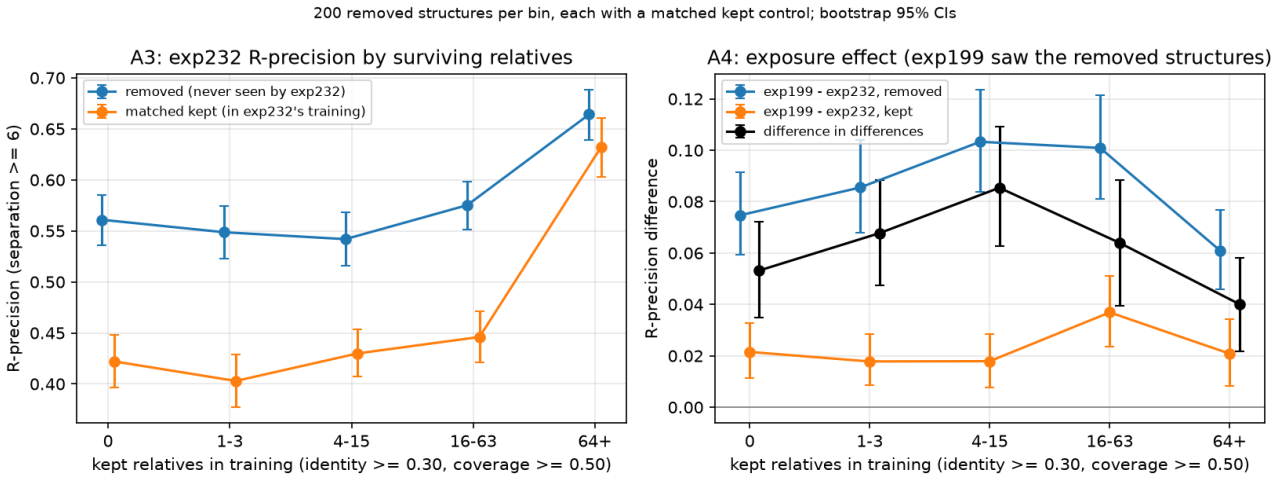
#225 removed 166,679 AFDB structures (4.04% of the corpus) because they resembled eval proteins. From 300 of the 2,067 AFDB shards we took 23,745 removed and 12,000 kept candidates, counted each one's relatives in exp232's 69.5M training sequences (MMseqs2, identity >= 0.30, coverage >= 0.50), and selected 200 removed structures per bin (0, 1-3, 4-15, 16-63, 64+ relatives), each matched one-to-one to a kept structure from the same bin on length, pLDDT and relative count (every standardized mean difference <= 0.08). Ground truth is each structure's own contacts-v1 document. Matching is within bin only: median removed length falls from 540 in bin 0 to 138 in 64+, and median pLDDT rises from 78.5 to 90.1, so bins are not comparable to each other. Details and paths are in `README.md`.

A4: exposure adds accuracy in every bin

Difference in differences: (exp199 − exp232 on removed) − (exp199 − exp232 on kept). The kept term absorbs how the two models differ overall. CIs bootstrap pairs within bin (10,000 resamples).

Relatives	exp199 − exp232, removed	exp199 − exp232, kept	DiD [95% CI]
0	0.075	0.021	0.053 [0.035, 0.072]
1-3	0.086	0.018	0.068 [0.047, 0.089]
4-15	0.103	0.018	0.085 [0.063, 0.109]
16-63	0.101	0.037	0.064 [0.040, 0.088]
64+	0.061	0.021	0.040 [0.022, 0.058]

Surviving relatives do not substitute for exposure: the effect at 64+ is smaller but still excludes zero. #225 purged whole clusters, so this measures exposure to a structure and its removed relatives, about 16 AFDB epochs of them. For scale, exp199's promotion over exp166 was +0.026, and #245 could not resolve the same model comparison on eval proteins (+0.024 [−0.022, +0.073]).



A3: no dose-response, and the control is not a ceiling

exp232 R-precision on removed structures it never saw:

Relatives	Removed	Matched kept	Removed − kept [95% CI]
0	0.561	0.422	0.139 [0.104, 0.173]
1-3	0.549	0.403	0.146 [0.108, 0.183]
4-15	0.542	0.430	0.112 [0.077, 0.147]
16-63	0.575	0.446	0.129 [0.094, 0.165]
64+	0.664	0.632	0.032 [−0.002, 0.066]

Structures with no surviving relatives score as well as those with 16-63. The rise at 64+ also appears in the kept controls, where proteins are much shorter. A piecewise-linear fit puts the saturation breakpoint at 7 relatives with a bootstrap CI of 2 to 435, so the curve gives no cap value.

The plan used kept structures as each bin's ceiling, but exp232 scores removed structures it never saw 0.11 to 0.15 above kept structures it trained on. Two explanations were tested and failed (`data/removed_gap_checks.json`). Similarity to eval proteins goes the wrong way: R-precision falls from 0.609 to 0.541 between the lowest and highest quartiles of identity to the nearest eval reference. Contact density and long-range share account for 0.009 of the 0.112 mean gap. The gap is unexplained, and until it is, the kept set is not a counterfactual for the removed set.

A2: the contact Jacobian

For every residue k and four substitutions (Ala, Gly, Pro, a charge flip), the production pairwise readout is rerun and the change in P(contact) is recorded for the protein's top-L predicted contacts (separation >= 6). P(contact) is the probability that {i, j} is the next contact statement, so it sums to about 1 over pairs; top-L values are around 0.004. Per contact:

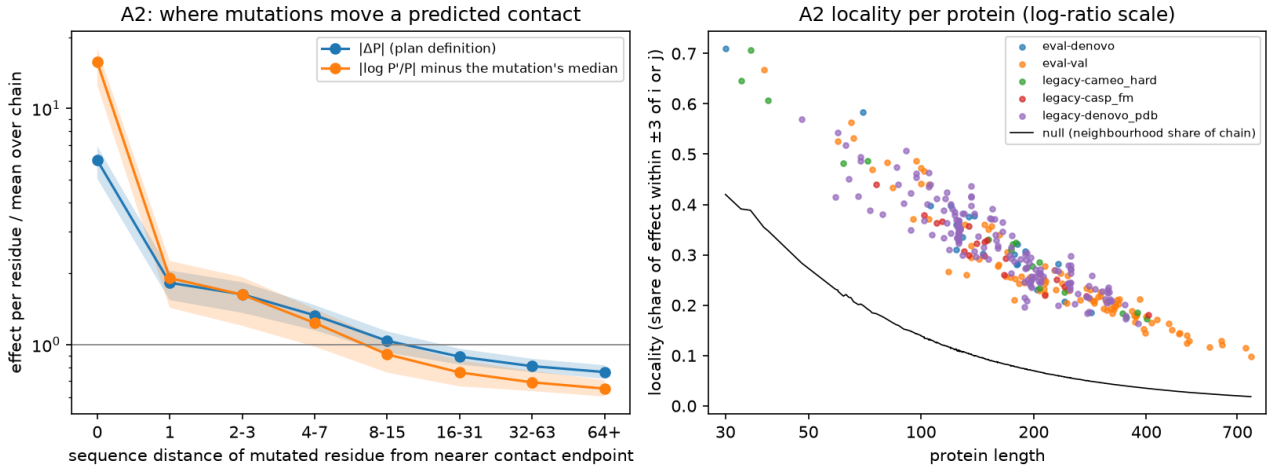
- **Locality**: the share of Σ|ΔP| from mutations within ±3 of i or j. The null, |N(i) ∪ N(j)| / L, is the share a mutation effect spread evenly over the chain would give.

- **Log scale**: the same share computed on |log(P′/P) − m|, where m is the mutation's median log ratio over the top-L contacts. Subtracting m removes the shift every contact takes when renormalization moves probability elsewhere. P′ is floored at 1e-5: knockouts at a contact's own residues overshoot to P′ as low as −1.1e-5 in fp32, and without a floor those strongest effects became NaN and were dropped.
- **Structural share**: the share from true-contact partners of i or j outside the ±3 window.

Eval set (exp277, bf16)

Across 316 proteins, locality is 0.193 [0.185, 0.202] against a null of 0.091 (log scale 0.302 [0.291, 0.314]). Per residue, a mutation at i or j itself moves the contact 6.0 times as much as the chain average (15.7 times on the log scale), the ±1 neighbours 1.8 times, and residues 64 or more away 0.77 times. The structural share, 0.028, barely exceeds its null of 0.023: mutating a residue that touches i or j in the structure moves the contact little more than mutating any other residue. Correct and incorrect predicted contacts have the same locality (difference 0.002 [0.000, 0.004]). The 37 novel-fold proteins with no training homolog sit at 0.189, the same as the rest.

These numbers are lower bounds. On 9 proteins of length 70 to 130 rerun in fp32, locality averaged 0.374 against 0.278 in bf16. Far from the contact, the bf16-minus-fp32 difference in ΔP has a median 0.6 to 5.2 times the median fp32 ΔP itself, depending on the protein. Because that error varies by protein, the covariate regression on this run is not interpretable: any coefficient, including length (−0.90 on logit locality), can absorb it, and the precision coefficient flips sign between scales (−0.049 raw, +0.317 log). An fp32 rerun of the 316 proteins would cost about 33 L40S-h; it has not been run.



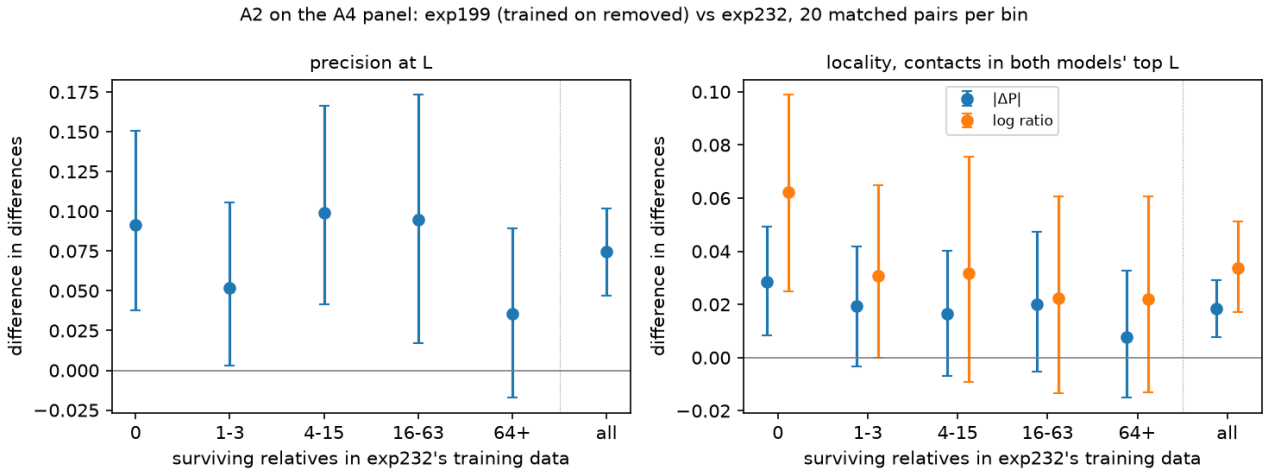
A4 panel (exp199 vs exp232, fp32)

The plan's test for recall: if exp199's advantage on the structures it trained on comes with a less local Jacobian than exp232's, while the kept controls show no difference, the advantage is recall. The subset is 20 matched pairs per bin with both members at most 250 residues (bin 0 has exactly 20 such pairs, well below its median length of 546), run in fp32.

Metric	exp199 − exp232, removed	exp199 − exp232, kept	DiD [95% CI]
Precision at L	0.080	0.005	0.075 [0.047, 0.102]
Locality, raw	0.016	0.001	0.015 [0.005, 0.024]
Locality, log	0.025	−0.003	0.028 [0.014, 0.044]
Locality on shared contacts, raw	0.020	0.001	0.018 [0.008, 0.029]
Locality on shared contacts, log	0.031	−0.003	0.034 [0.017, 0.051]
Structural share, raw	0.002	−0.001	0.003 [0.001, 0.004]

Exposure raised precision at L, reproducing A4 on this subset, and raised locality instead of lowering it. "Shared contacts" restricts both models to contacts in both of their top-L sets, so the increase is a change in how exp199 computes the same contacts, not in which contacts it predicts. Per bin, every locality DiD is positive; with 20 pairs, only bin 0 excludes zero on its own. In fp32 the structural share is 1.4 to 1.9 times its null (0.031 to 0.036 against 0.019 to 0.023), and exposure moves it by 0.003.

Locality and precision move together. In a check outside the committed outputs, their Spearman correlation across the 400 structure-model rows is 0.53 (log scale), and the 23 pairs whose precision DiD is under 0.02 in magnitude have a locality DiD of 0.001. The data do not say whether exposure sharpens locality directly or whether locality follows from a more confident prediction.



What this means for decision A

The plan read high locality as coevolution-like coupling and dependence spread over the chain as recall. In exp199, training on a structure moved dependence toward the contact's own residues. One reading that fits, not tested here, is memorization of local sequence at both

endpoints (the motif-pair hypothesis in Zhang et al. 2024): a ±3 window cannot tell that apart from coupling. Either way, a high locality on the eval set does not show the model avoids recall, and the locality-at-matched-MSA-depth test in section 5.2 cannot decide A.

That leaves A4 as the evidence. exp199 carries a memorized component worth 0.04 to 0.085 R-precision on structures it trained on for about 16 epochs, present at every relative count. A per-cluster cap would remove some of that; the panel does not say what the cap should be or what it would cost on unseen proteins, because A3 has no working control and its curve is flat. A cap ablation in training, or an A3 control matched on whatever makes removed structures easier, would supply the number.

Limits

- The eval-set A2 run and the panel A2 run use different models (exp277 vs exp199/exp232) and precisions (bf16 vs fp32). Their absolute localities are not comparable.
- A2 used one seed per protein; the R-precision readouts used four.
- The panel A2 subset is the short tail of each bin, and its precision is precision at L, not R-precision.
- exp199 and exp232 differ in training length and schedule as well as corpus; the kept-set term removes overall differences, not interactions between schedule and protein type.
- Not run: A1 (needs a gradient path through the readout), the A4 document-level "seen vs fresh offset" variant, harness checks 3 and 4, and the fp32 rerun of the eval-set A2.

Files

Data	Script
data/contamination_did.csv, data/relative_dose_response.csv, data/relative_dose_fit.json, data/panel_scores/	analyze_panel.py, plot_panel.py
data/a2_per_protein.csv, data/a2_distance_profile.csv, data/a2_summary.json, data/a2_precision_check.json, data/a2_timings.csv	a2_contact_jacobian.py, analyze_a2.py, plot_a2.py
data/a2_panel_proteins.csv, data/a2_panel_per_structure.csv, data/a2_panel_did.csv	a2_panel_select.py, analyze_a2_panel.py, plot_a2_panel.py

Raw outputs: `s3://marinfold-exp91-usw2/exp341/` (`panel/`, `a2/`, `relatives/`).